

C2Explainer: Customizable Mask-based Counterfactual Explanation for Graph Neural Networks

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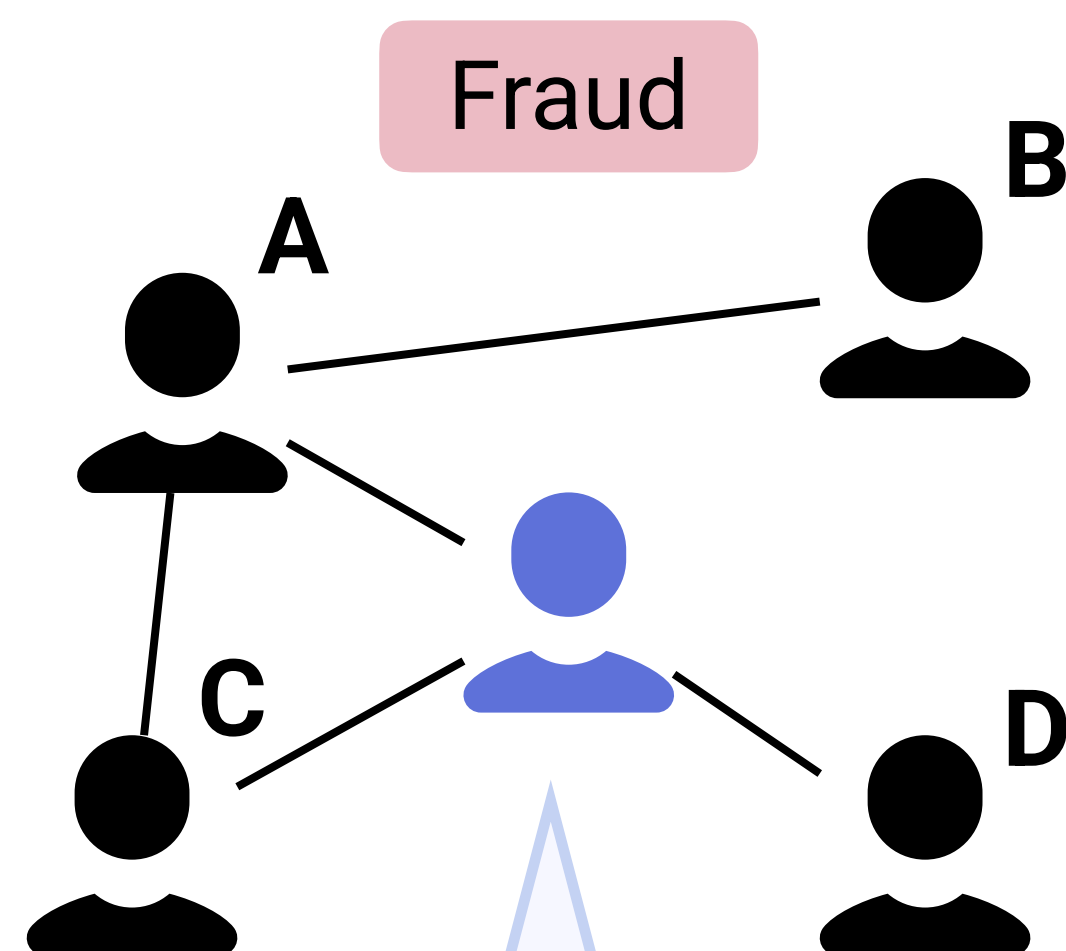
Kyoto University

Motivation

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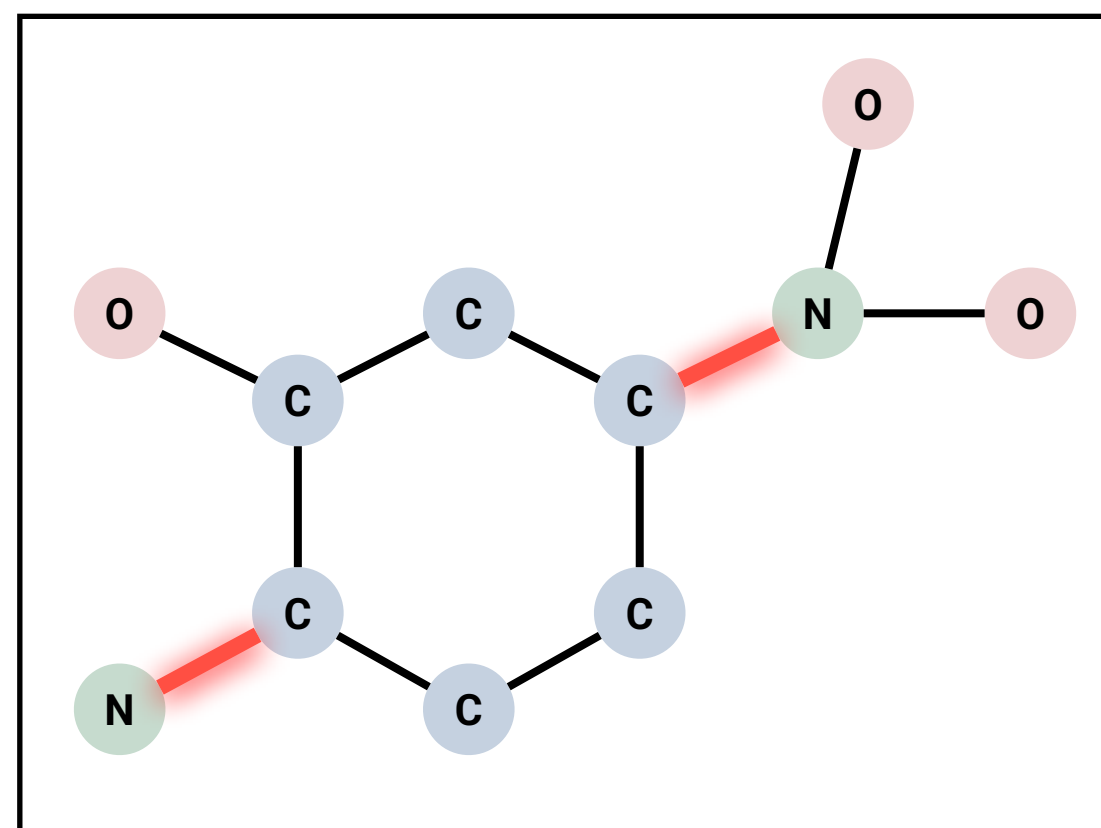
- **Graph Neural Networks** have achieved wide success across diverse fields
- How can we **trust the predictions** of GNNs?

Fraud detection



Why was my account flagged as fraud?
What steps can I take to reverse the decision?

Molecule Classification



Adding or removing which edge may change the model's prediction?

Providing answers to these questions improves model trustworthiness.

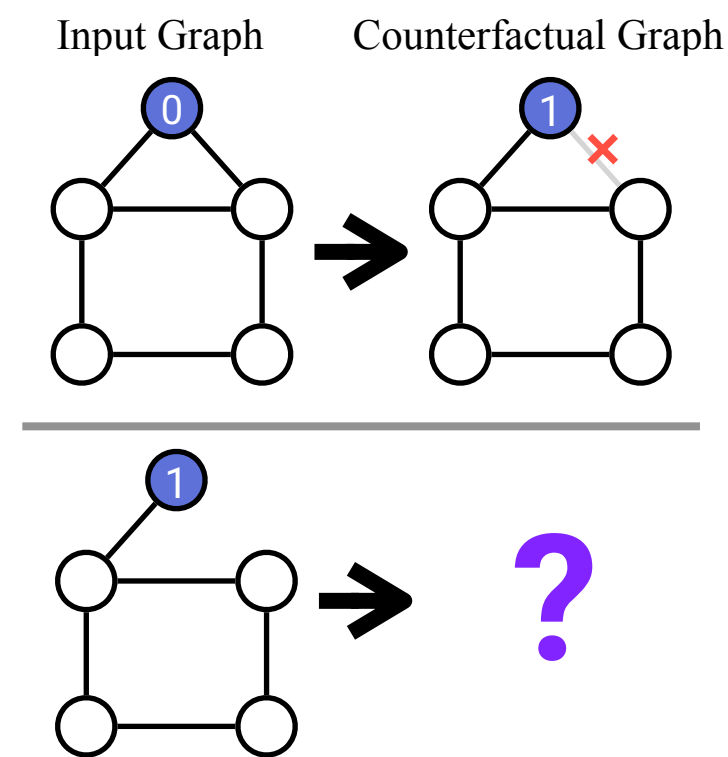
Counterfactual explanations (CE) are designed to address these questions.

Our contributions

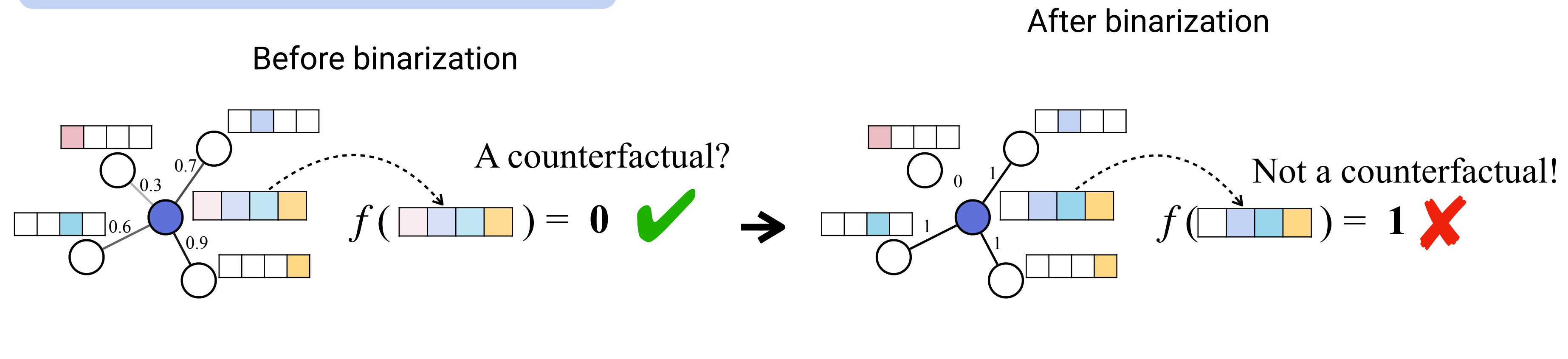
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- We point out two limitations of previous counterfactual explanation (CE) methods:

1. Cannot add edges



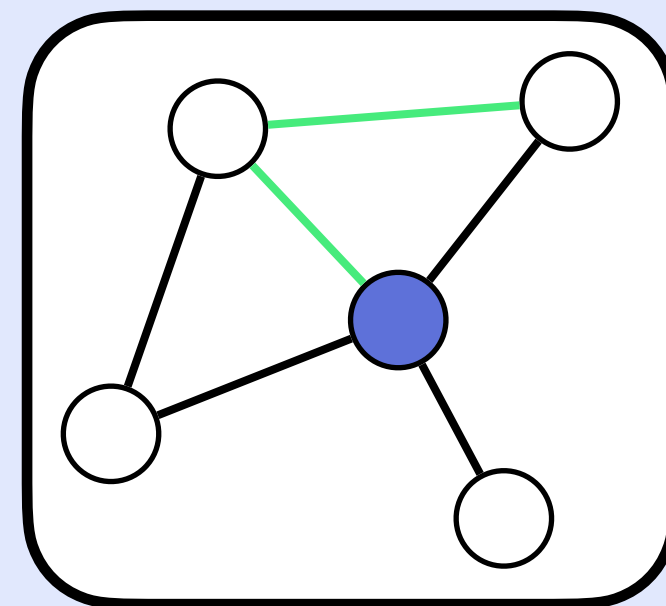
2. Introduced evidence problem



- We propose a new framework **C2Explainer** to address these limitations and to provide high quality explanations

Supergraph as input

enable **Edge Addition**



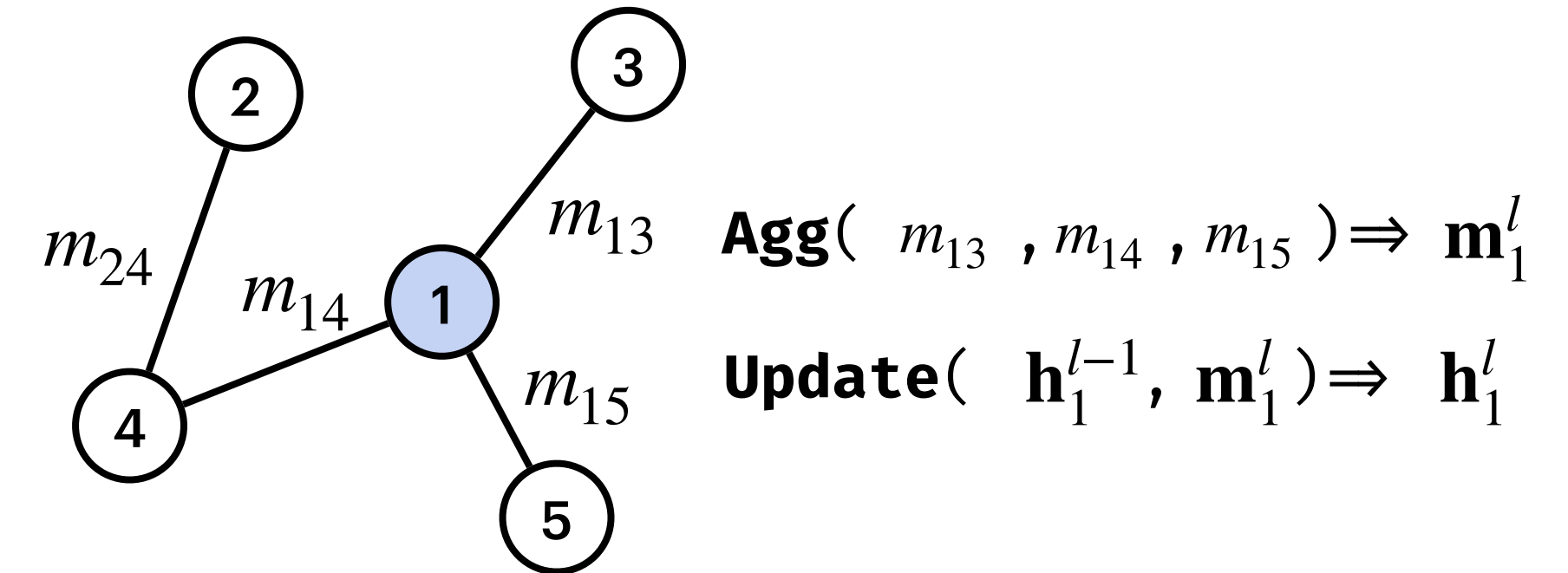
Straight-through unit

- X** continuous relaxation
- ✓** retain **binary** edge mask adjacency matrix

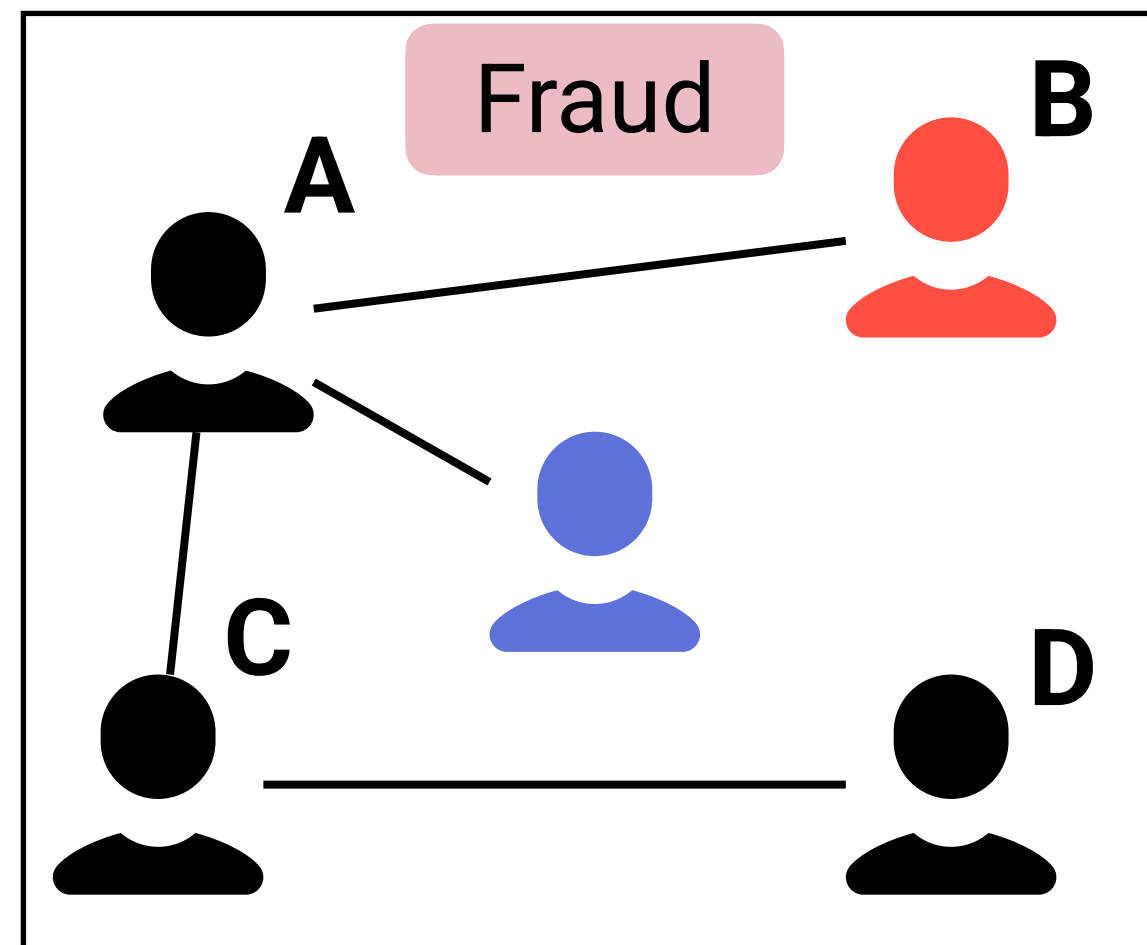
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Preliminaries

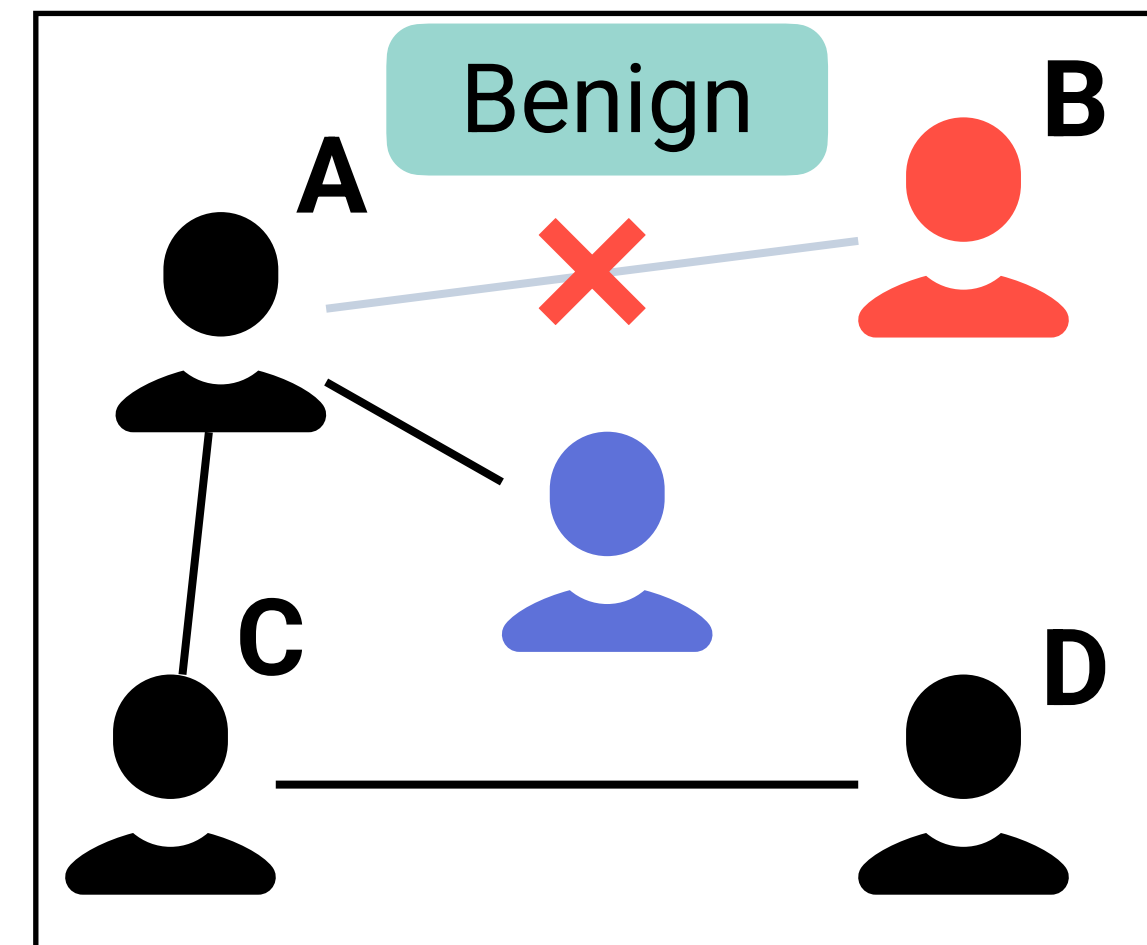
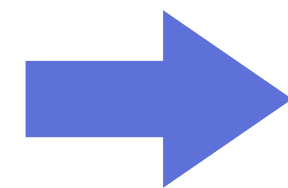
- **Graph Neural Networks:** deep learning models for graph structured data.
- Most GNNs are **Message-Passing Neural Networks**



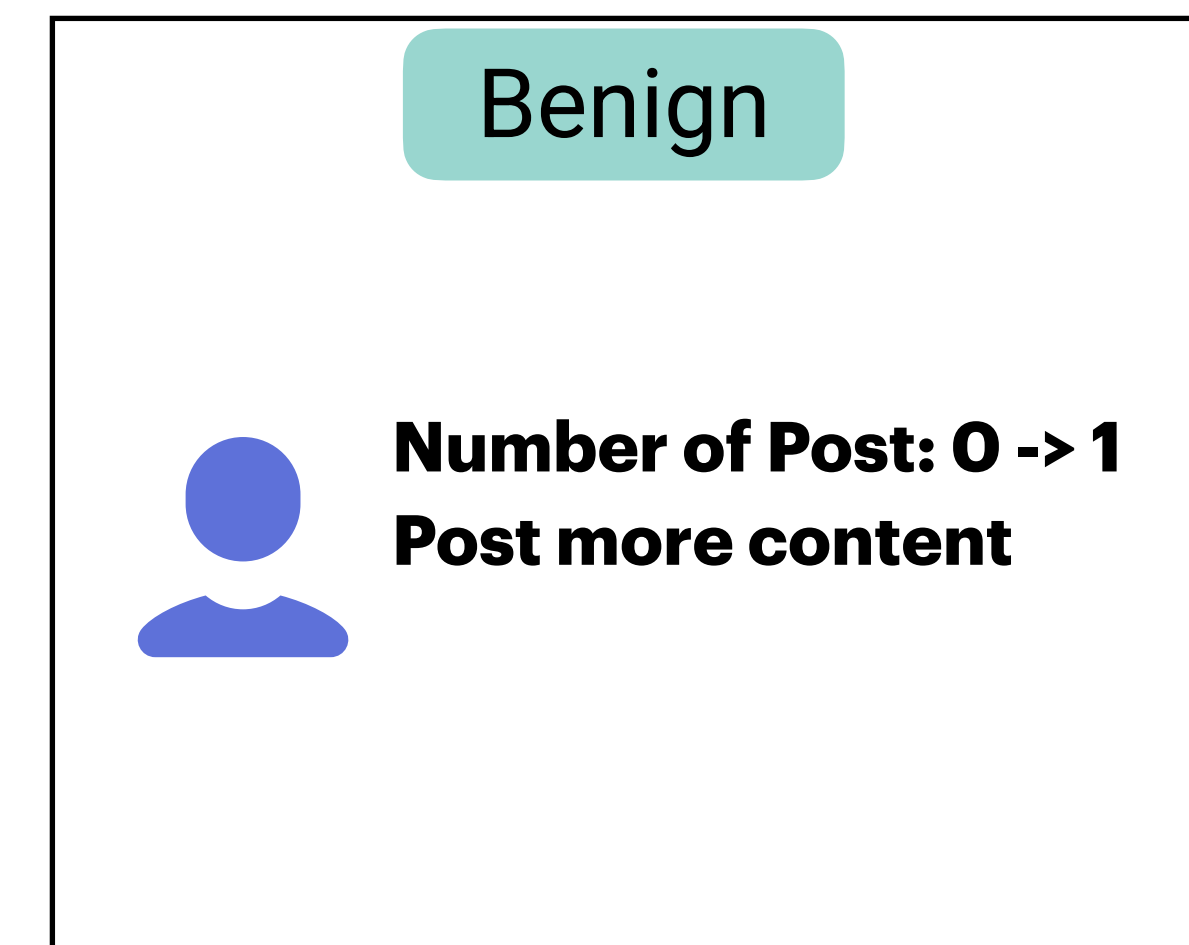
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- **Counterfactual Explanation:**



Social Network



Counterfactual 1



Counterfactual 2

CEs for GNNs: difficulty

Tabular data

- Finding a counterfactual can be formulated as an optimization problem:

$$\mathbf{x}^{\text{CF}} = \arg \min_{\mathbf{x}'} \text{dist}(\mathbf{x}', \mathbf{x}) \quad \text{s.t.} \quad f(\mathbf{x}') \neq f(\mathbf{x})$$

Where $\text{dist}(\cdot, \cdot)$ is a function measures the dissimilarity between two input instances, and f represents the prediction model.

- Can be optimized the same way as model training:

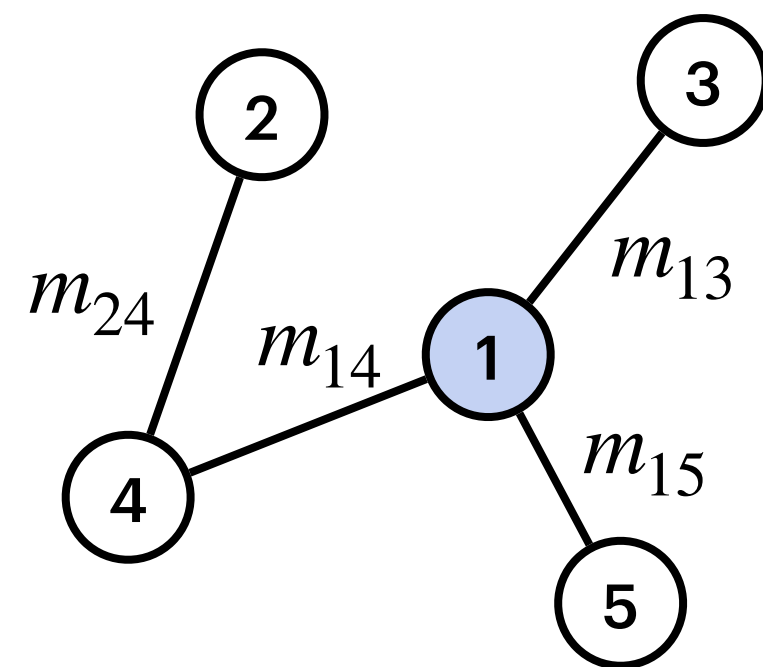
$$\arg \min_{x'} \mathcal{L} \left(f_w(x'), y^{\text{desired}} \right) + \lambda \text{dist}(x', x),$$

where y^{desired} is a label different from the original prediction.

Graph data

The adjacency matrix $A \in \{0,1\}^{n \times n}$ of a graph is **binary**

- Prior studies (GNNExplainer_[1], CF-GNNExplainer_[2], CFF_[3], AMExplainer_[4], PGExplainer_[5]) address this problem by **relaxing the adjacency matrix from binary to continuous**.
- Equivalent to introducing an **fractional edge mask** $M \in [0,1]^{n \times n}$.



$$\text{Agg}(m_{13}, m_{14}, m_{15}) \Rightarrow \mathbf{m}_1^l$$

$$\text{Update}(\mathbf{h}_1^{l-1}, \mathbf{m}_1^l) \Rightarrow \mathbf{h}_1^l$$



Message Mask

$$\text{Agg}(m_{13} \cdot M_{13}, m_{14} \cdot M_{14}, m_{15} \cdot M_{15}) \Rightarrow \mathbf{m}_1^l$$

$$\text{Update}(\mathbf{h}_1^{l-1}, \mathbf{m}_1^l) \Rightarrow \mathbf{h}_1^l$$

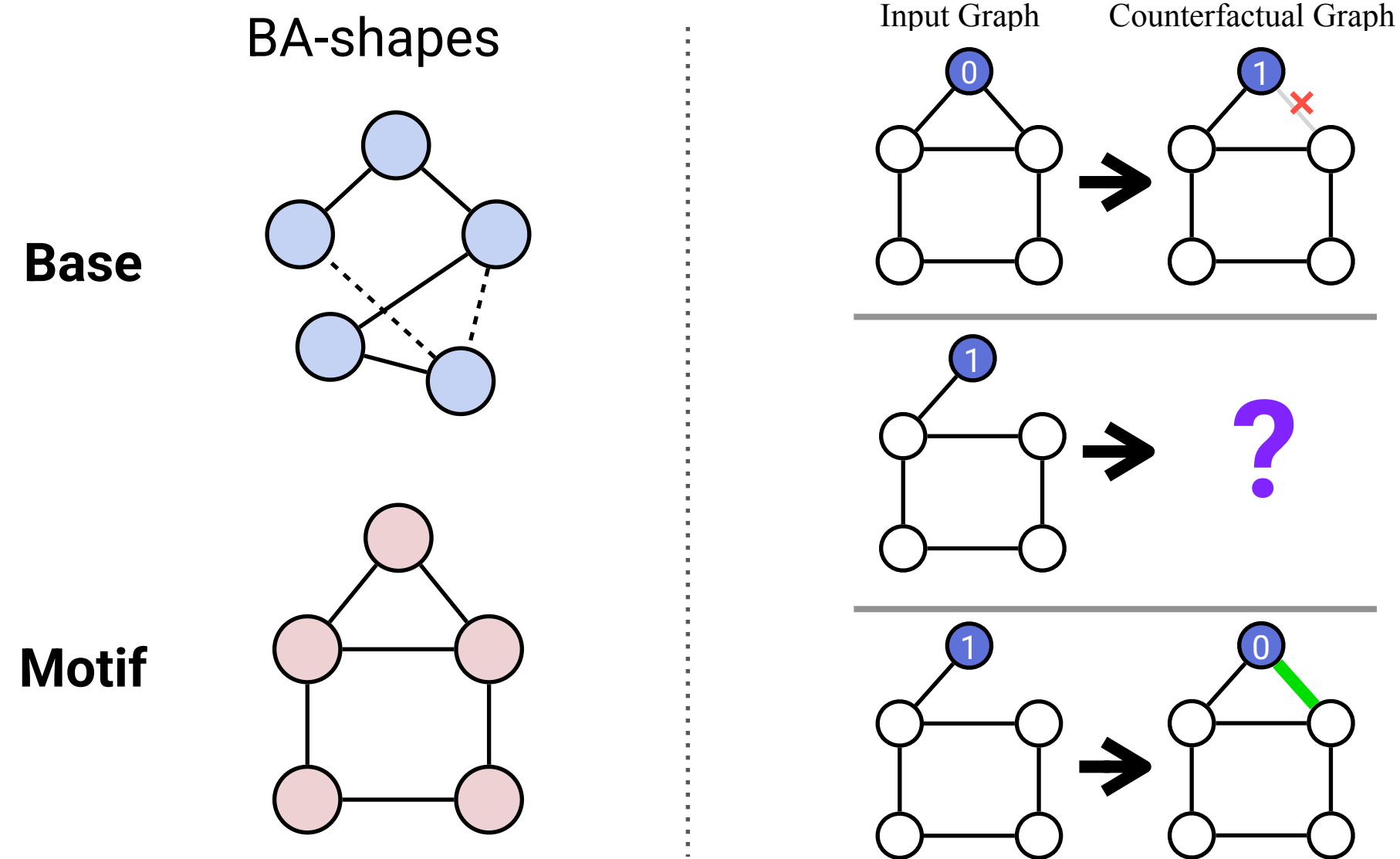
1. Ying, Zhitaoy, et al. "Gnnexplainer: Generating explanations for graph neural networks." *NeurIPS* 2019.
 2. Lucic, Ana, et al. "Cf-gnnexplainer: Counterfactual explanations for graph neural networks." *AISTATS* 2022.
 3. Tan, Juntao, et al. "Learning and evaluating graph neural network explanations based on counterfactual and factual reasoning." *WWW* 2022.

4. Zhang, Wei, Xiaofan Li, and Wolfgang Nejdl. "Adversarial Mask Explainer for Graph Neural Networks." *WWW* 2024.
 5. Luo, Dongsheng, et al. "Parameterized explainer for graph neural network." *NeurIPS* 2020.

Limitations of previous works

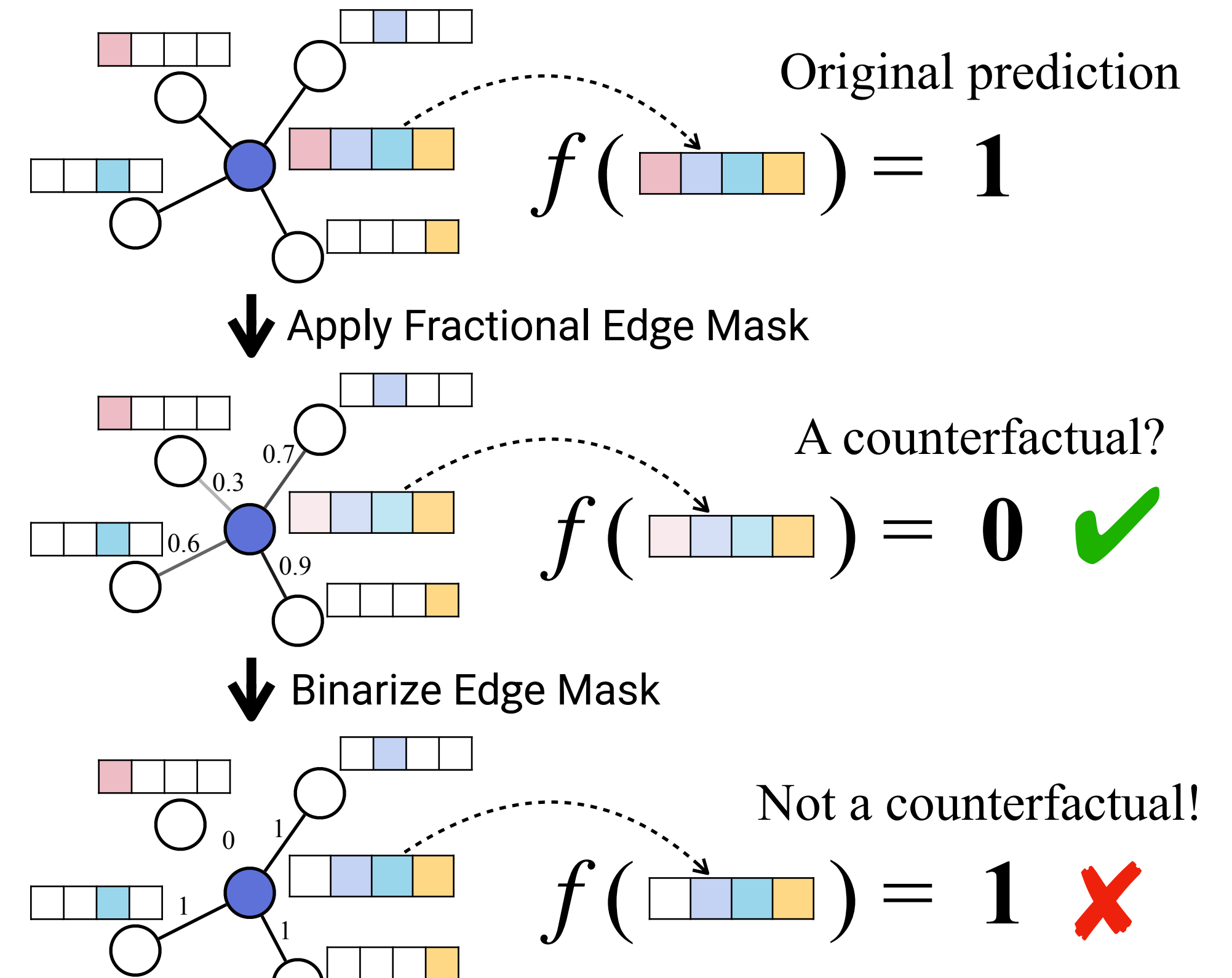
Cannot add edges

- Edge mask can only mask out existing edges, meaning that 0 values in the adjacency matrix are unaffected.
- Suitable for identifying critical subgraph structures but is inappropriate for generating counterfactuals.



"Introduced evidence" problem

- The fractional edge mask introduces **new semantic meaning** to the node embeddings, which the prediction model has not encountered during training, thus affecting the explanation quality.



1. Ying, Zitao, et al. "Gnnexplainer: Generating explanations for graph neural networks." *NeurIPS* 2019.
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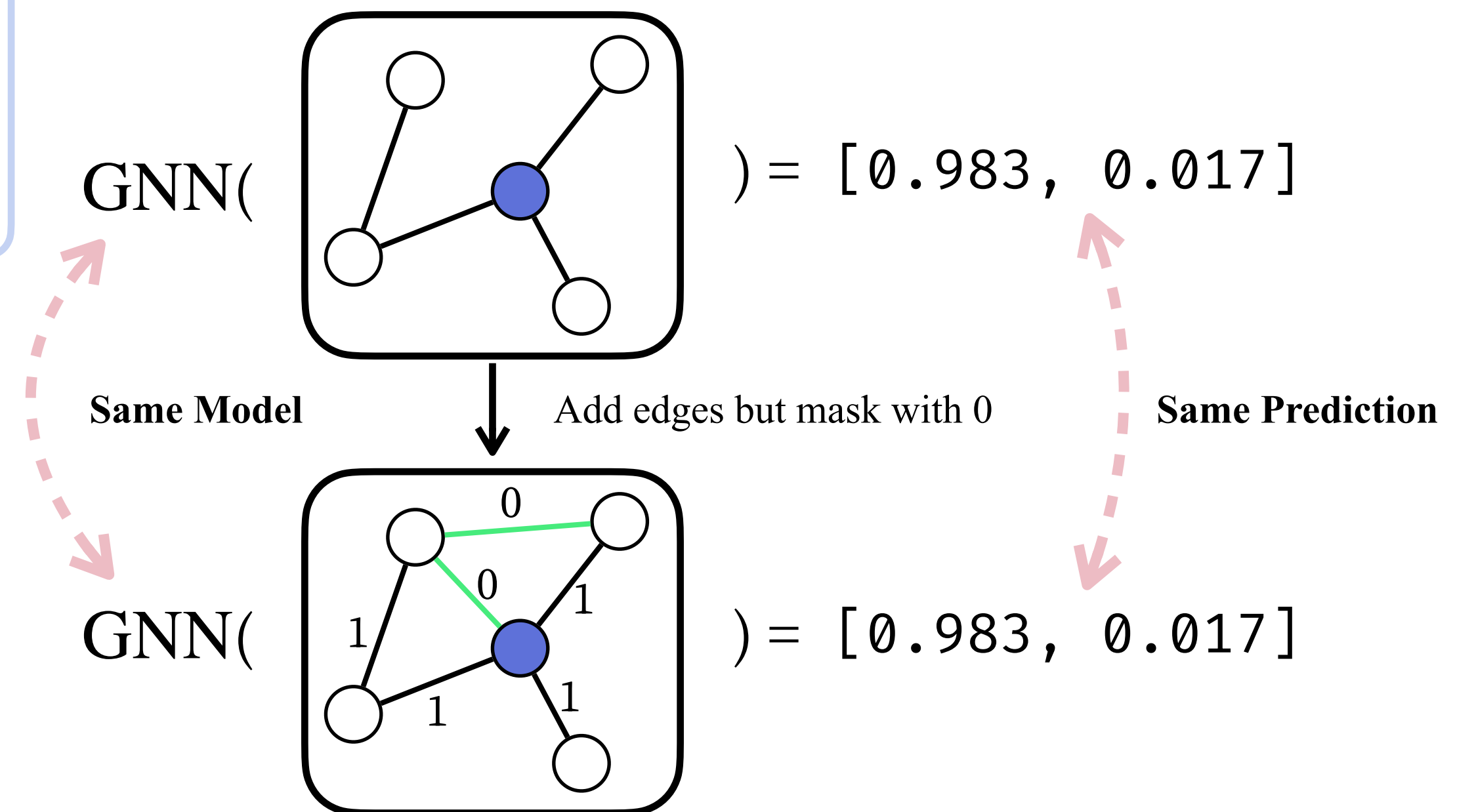
Supergraph as input

Observation: The GNN model gives the same output whether using the original graph or a graph with extra edges masked (mask=0 for extra edges, 1 for original edges).

- **C2Explainer:**

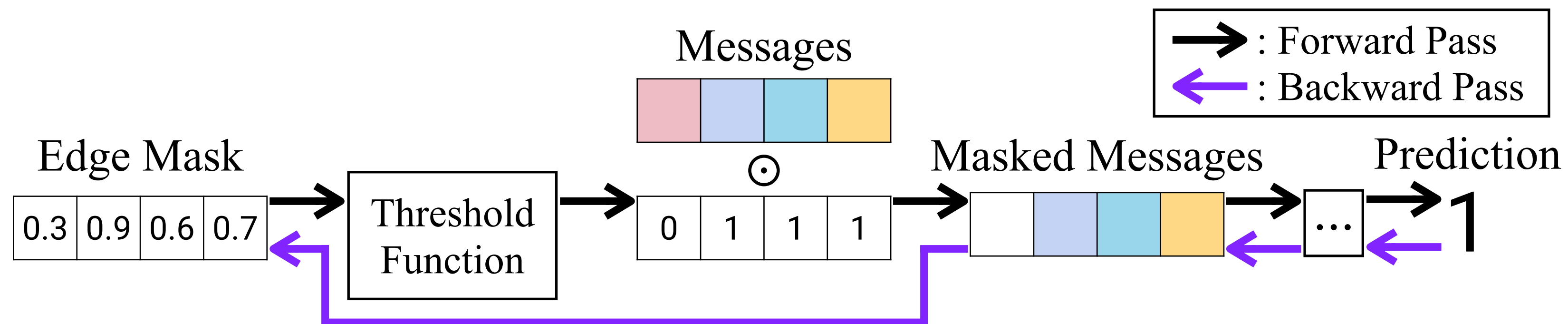
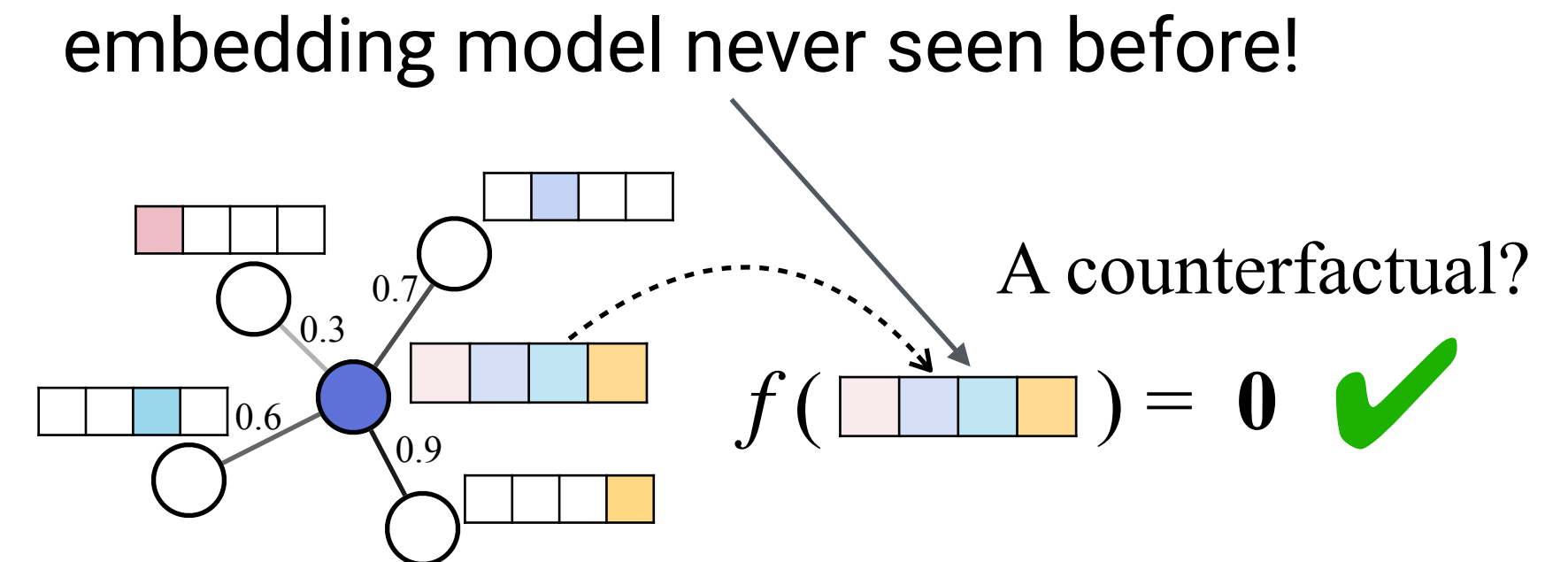
- First, finds possible edges that could affect the target node's embedding.
Using a complete graph as input or directly perturbing a dense adjacency matrix is often computationally infeasible.
- Then, samples a subset of extra edges for explanation.

- Implemented using a sparse edge list instead of a dense adjacency matrix, ensuring computational efficiency.



Straight-thought unit

- **Previous works:** add a regularization term (element-wise entropy) to loss function to penalize mask values that are not close to 0 or 1.



- During the forward pass, we binarize the mask values to ensure the edge mask contains only 0 or 1 when computing model predictions.
- During the backward pass, we use the soft edge mask values to compute gradients.
- **Result:** Semantic meaning is preserved during optimization

Customizable Masks

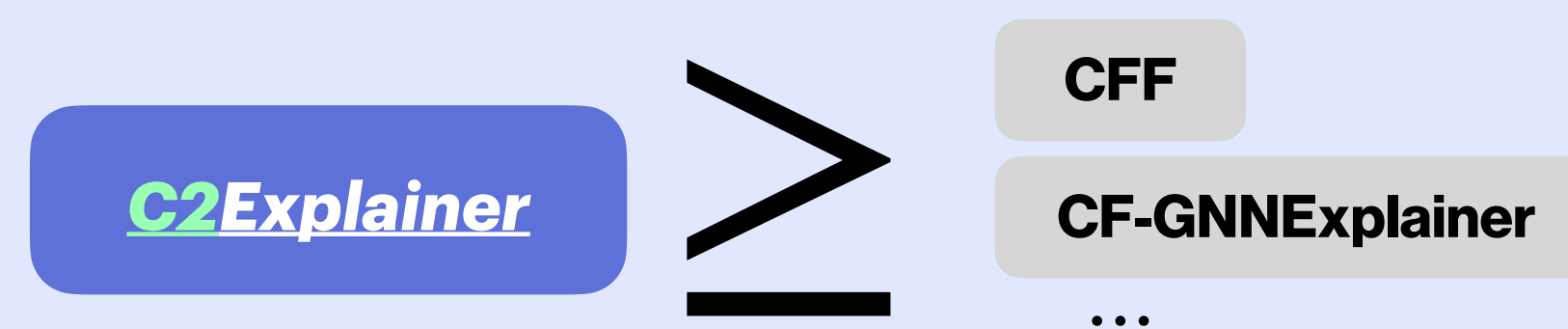
- To accommodate diverse user needs and constraints across various domain-specific tasks, we offer multiple customization options for the edge mask and feature perturbation matrix.
- **Edge mask:**
 - specify which edges should not be perturbed
 - whether allowing add edges
- We build an **algorithmic recourse (AR) mode** and **node explanation mode** by customizing masks!

Experiments

CF validity



CF #perturbs & Similarity



Model	Tree-cycles			Tree-grid			BA-shapes		
	Validity $\uparrow$	#perturbs $\downarrow$	Similarity $\uparrow$	Validity $\uparrow$	#perturbs $\downarrow$	Similarity $\uparrow$	Validity $\uparrow$	#perturbs $\downarrow$	Similarity $\uparrow$
GNNExplainer	0.368	4.531	0.549	0.581	7.965	0.573	0.343	4.854	0.778
PGExplainer	0.397	3.797	0.646	0.581	4.734	0.739	0.350	4.388	0.780
CFF	0.425	5.068	0.536	0.589	8.228	0.554	0.529	4.378	0.827
CF-GNNExplainer	0.678	1.703	0.866	0.671	1.545	0.925	0.443	1.935	0.957
Ours(subgraph)	0.724	1.278	0.894	0.695	1.462	0.931	0.600	1.845	0.958
Ours	0.994	1.572	0.827	0.984	1.326	0.908	0.929	2.177	0.956

Model	Loan-decision			BA2motifs			MUTAG		
	Validity $\uparrow$	#perturbs $\downarrow$	Similarity $\uparrow$	Validity $\uparrow$	#perturbs $\downarrow$	Similarity $\uparrow$	Validity $\uparrow$	#perturbs $\downarrow$	Similarity $\uparrow$
GNNExplainer	0.005	5.000	0.968	0.270	3.000	0.882	0.368	2.929	0.874
PGExplainer	0.010	4.500	0.955	0.245	2.041	0.861	0.474	3.000	0.875
CFF	0.165	4.909	0.975	0.200	2.925	0.887	0.474	3.000	0.875
CF-GNNExplainer	0.260	1.769	0.987	0.775	2.858	0.888	0.553	2.048	0.915
Ours(subgraph)	0.335	2.164	0.981	0.940	1.803	0.930	0.868	2.121	0.880
Ours	0.930	1.204	0.993	0.950	1.784	0.930	1.000	1.842	0.884

Use Cases

Algorithmic Recourse (AR)

- **AR: Counterfactual explanations with actionability**
- AR not only provides explanations but also offers recommendations for end-users to follow, helping them reverse undesired model decisions.

By customizing the masks, C2Explainer can always provide actionable recommendations to end-users.

Model	Validity	#Perturbs	Similarity	AR Val.
CF-GNNExplainer	0.260	1.769	0.987	0.000
Ours(AR mode off)	0.930	1.204	0.993	0.730
Ours(AR mode <i>M</i>)	0.905	1.425	0.992	0.905
Ours(AR mode <i>M+P</i>)	0.990	0.917	0.994	0.990

Bias Detection

Recourse bias:

We trained two models

- Model 1: unbiased
- Model 2: biased, only see female with label 0 while training.

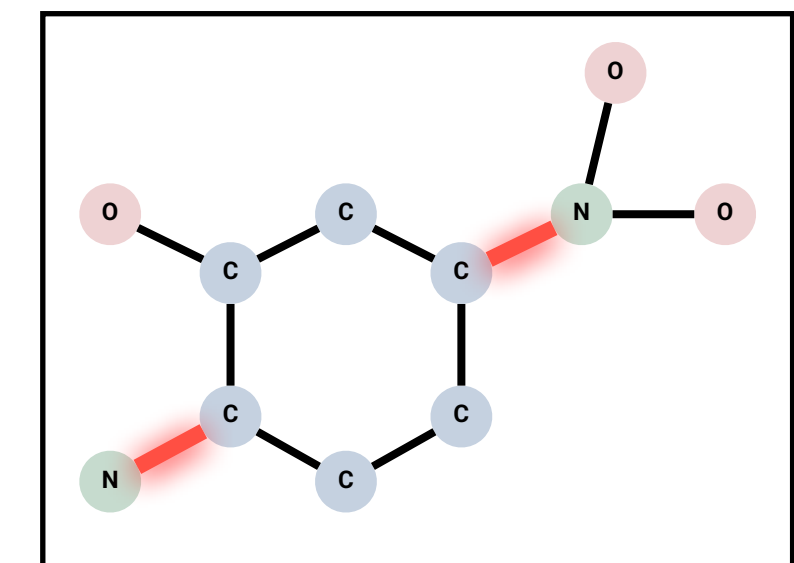
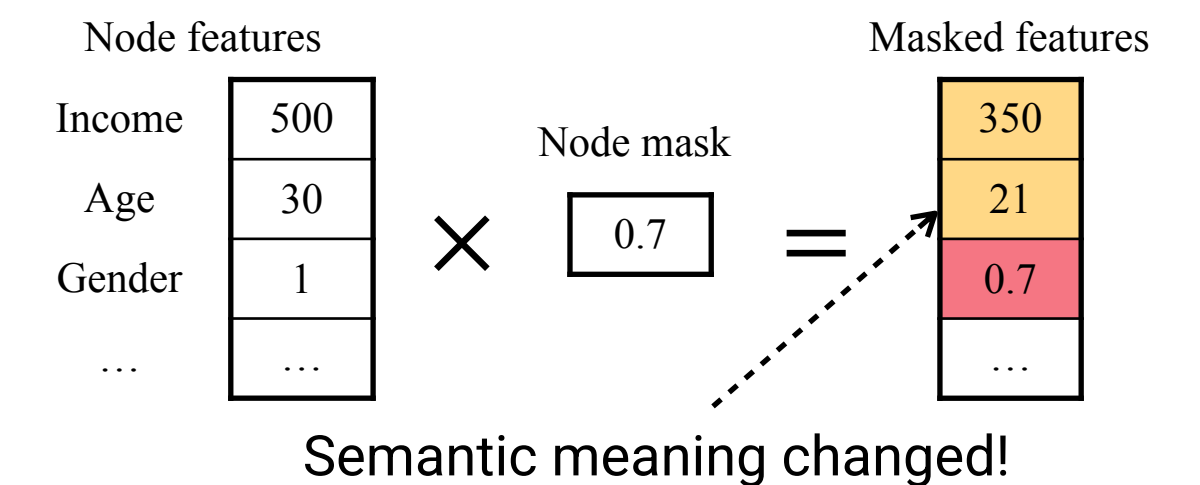
C2Explainer successfully identified recourse bias against female on model 2.

Table: Recourse bias detection for classification models

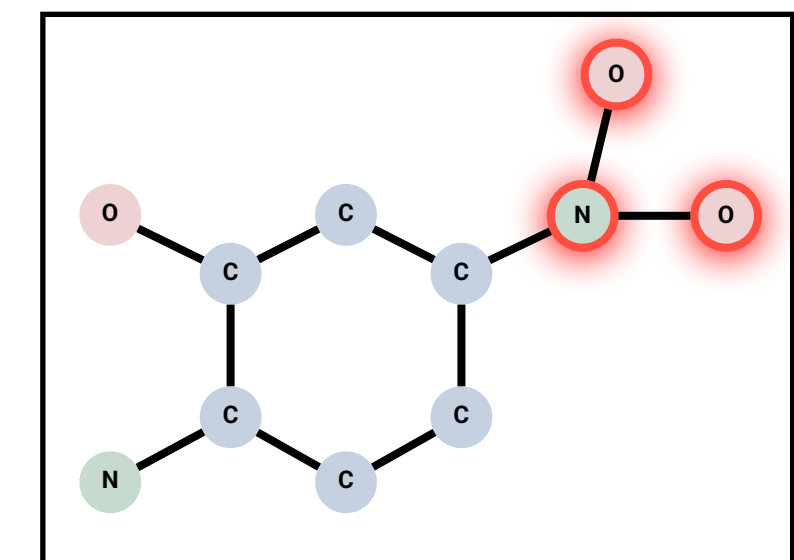
Model	Gender	AR Validity	#Perturbs	Similarity
Model 1	Male	0.788	1.441	0.993
	Female	0.792	1.754	0.988
	Diff	-0.004	0.313	0.005
Model 2	Male	0.729	1.593	0.992
	Female	0.573	2.745	0.983
	Diff	0.156	-1.153	0.009

Node-level Explanation

Figure: Conventional node mask



(a) edge level explanation



(b) node level explanation

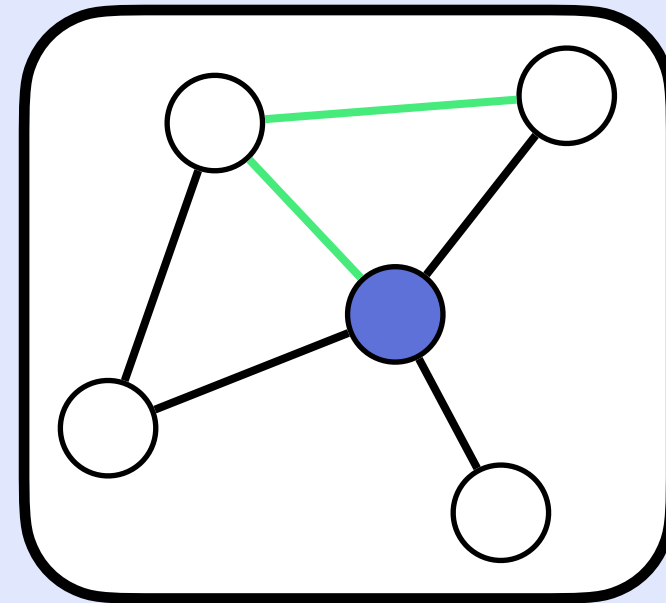
Conclusion

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- We point out two limitations of existing CE methods: 1. Cannot add edges 2. Introduced evidence problem
- We propose a new framework **C2Explainer** to address these limitations and to provide high quality explanations

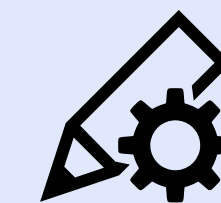
Supergraph as input

enable **Edge Addition**



Straight-through unit

- ✗ continuous relaxation
- ✓ retain **binary** edge mask adjacency matrix



**Customizable
Masks**

...

- We conduct extensive experiments and showcase real-world applications
- Future work: improve computational efficiency and design better evaluation benchmarks.

Thanks for your listening!